



Memo version 2.0

BESIII Analysis Memo

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BESIII Analysis Memo (March 2, 2026)

Cross section measurement of process $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

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Abstract

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List of changes

For an easy reading of the document, major changes in various versions with respect to the previous one are listed below.

Version 1.0

- First version of memo.

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1 Introduction

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2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider—is operating in the τ -charm energy region (2 to 4.9 GeV) with the design luminosity of $1 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$ [1]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric e^+e^- collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the e^+e^- interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [2]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit dE/dx in MDC and flight time in TOF and forms likelihoods $\mathcal{L}(h)(h = K, \pi)$ for a hadron h hypothesis. More details about the BESIII detector are provided in Ref [3].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [4, 5].

$E_{\text{beam}}(\text{GeV})$	Integrated luminosity (pb^{-1})	Data taking time	RunNo
3.773	$2931.8 \pm 0.2 \pm 13.8$ [4]	2010(round03) 2011(round04)	11414 – 13988, 14395 – 14604 20448 – 23454
3.773	4995.0 ± 19.0 [5]	2022(round15)	70522 – 73929
3.773	8157.0 ± 31.0 [5]	2023(round16)	74031 – 78536
3.773	4191.0 ± 16.0 [5]	2024(round17)	78615 – 81094

Table 2: Processes and luminosity scales of the inclusive MC samples used in this analysis. Separate samples are available for each data-taking round. Most processes use samples with an integrated luminosity equivalent to ten times that of the data.

Process	Luminosity scale
$e^+e^- \rightarrow \psi(3770) \rightarrow D^0\bar{D}^0$	$10 \times$
$e^+e^- \rightarrow \psi(3770) \rightarrow D^+D^-$	$10 \times$
$e^+e^- \rightarrow \tau^+\tau^-$	$10 \times$
$e^+e^- \rightarrow \psi(3770) \rightarrow \text{non-}D\bar{D}$	$10 \times$
$e^+e^- \rightarrow q\bar{q}$	$10 \times$
$e^+e^- \rightarrow \gamma_{\text{ISR}}\psi(2S)$	$10 \times$
$e^+e^- \rightarrow \gamma_{\text{ISR}}J/\psi$	$10 \times$
$e^+e^- \rightarrow e^+e^-$	$0.25 \times$
$e^+e^- \rightarrow \gamma\gamma$	$3 \times$
$e^+e^- \rightarrow \mu^+\mu^-$	$15 \times$

3 Data and MC samples

We use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1} , collected by BESIII detector at center-of-mass energy of 3.773 GeV, under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Ref. [4, 5].

Simulated Monte Carlo (MC) samples are employed to study backgrounds, optimize event selection, and estimate detection efficiencies. Two categories of MC samples are utilized in this analysis:

- **The inclusive MC samples** comprehensively model known physics processes at $\sqrt{s} = 3.773 \text{ GeV}$, including $D\bar{D}$ production, $\tau^+\tau^-$ pair production, initial-state-radiation (ISR) production of vector charmonium states (J/ψ and $\psi(2S)$), QED processes, and continuum light hadron production. The charm working group provides separate samples for the different data-taking rounds(round03-04, round15, round16, and round17) [7]. In this analysis, we use inclusive MC samples with an integrated luminosity equivalent to ten times that of the data for most processes, except for the QED processes $e^+e^- \rightarrow e^+e^-$, $\gamma\gamma$, and $\mu^+\mu^-$, where we use the samples with the scaling factors provided by the charm working group. The processes and the corresponding luminosity scales of

the samples used in this analysis are summarized in Table 2.

- **Exclusive signal and background MC samples** generated specifically for this analysis using the PHOKHARA event generator [8, 9, 10] (version 9.1). These samples are produced independently for each data-taking round (round03, round04, round15, round16, round17) to account for variations in detector conditions. Sufficient statistics are generated for each process to ensure reliable efficiency determination and background studies. The generated processes include:

$$- e^+ e^- \rightarrow \pi^+ \pi^- \pi^0 \gamma_{\text{ISR}} \text{ (signal process)}$$

$$- e^+ e^- \rightarrow \pi^+ \pi^- \pi^0 \pi^0$$

$$- e^+ e^- \rightarrow \pi^+ \pi^- \pi^0 \pi^0 \gamma_{\text{ISR}}$$

These exclusive samples are used for signal efficiency studies, background characterization, and validation of the analysis methodology.

4 Event selections

The process under investigation is $e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\text{ISR}}$, whose final states include two charged tracks, one neutral π^0 , and one ISR photon. The following selection criteria is applied to all the data samples. Here the figures are shown by taking $\sqrt{s} = 3.773\text{GeV}$ round16 data as an example. The distributions related to the event selection of other data sets are shown in Appendix. XXX. Situations are similar to each other.

- Good charged track: A charged track is taken as a good MDC track if it originates from the beam vertex and lies within the MDC polar angle region:
 - $|\cos\theta| < 0.93$, where θ is the polar angle with respect to the beam axis.
 - $|\Delta r| < 1\text{ cm}$, $|\Delta z| < 10\text{ cm}$, where $|\Delta r|$ and $|\Delta z|$ is distance of closest approach to the interaction point (IP) perpendicular to the beam axis and along the beam axis, respectively.
 - the ratio of the calorimeter-deposited energy to the track momentum E_{EMC}/p is required to be less than 0.8 for both of the charged tracks.
 - no PID requirements for π^+ , π^- .
- γ selection:
 - $E > 0.025\text{ GeV}$ (barrel: $|\cos\theta| < 0.8$), $E > 0.050\text{ GeV}$ (endcap: $0.86 < |\cos\theta| < 0.92$).
 - $0 \leq t_{\text{TDC}} \leq 14(\times 50\text{ns})$ to suppress fake photons due to electronic noise or beam background.
 - the closest angle with all charged tracks $\theta > 10^\circ$ to suppress background from radiation of charged tracks.
- $\pi^0 \rightarrow \gamma\gamma$ selection:
 - two photon invariant mass $M_{\gamma\gamma}$ within $(0.115, 0.150)\text{ GeV}/c^2$.
 - To improve the momentum resolution, 1C kinematic fit is performed. Here, the 1C means π^0 mass constraint, and the updated 4-momentum will be used in the further analysis. $\chi_{\text{KF}}^2 < 10$

Candidate events are selected by requiring exactly two good charged tracks, accompanied by at least one reconstructed π^0 and a minimum of three good photon showers. This ensures the presence of at least one π^0 and one isolated photon from Initial-State Radiation (γ_{ISR}). A four-constraint (4C) kinematic fit is then applied to the event, and the candidate with the smallest $\chi_{4\text{C}}^2$ value is chosen for further analysis.

After identifying an ISR photon from the isolated photon showers, the invariant mass of the remaining $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ system is reconstructed. As expected for ISR processes, the resulting center-of-mass energy $\sqrt{s'}$ of the $\pi^+\pi^-\pi^0$ system shows a continuous spectrum, characteristic of the radiative return mechanism. This continuum behavior is illustrated in Fig. 1 across the $\omega(782)$, $\phi(1020)$, and ω'/ω'' mass regions.

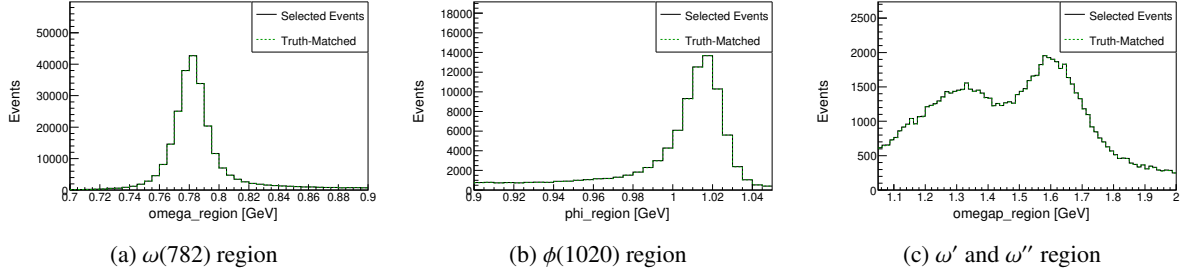


Figure 1: Invariant mass spectra of MC in the (a) $\omega(782)$, (b) $\phi(1020)$, and (c) ω'/ω'' mass regions. The distributions correspond to $M(pK_S^0)$, $M(p\pi^0)$, and $M(K_S^0\pi^0)$, respectively. The "Truth-Matched" distribution (green dashed line) corresponds to reconstructed events where the momentum directions of the π^0 and γ_{ISR} successfully match their truth-level counterparts. A loose requirement of $\chi_{\text{KF}}^2 < 100$ is applied to the kinematic fit. The continuous spectrum of $\sqrt{s}$ for the $\pi^+\pi^-\pi^0$ system demonstrates the radiative return mechanism via ISR.

5 Background Study

In the study of background processes, the main backgrounds are categorized into $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$, $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$, QED processes, and others. The dominant contributions arise from processes with additional π^0 , particularly $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$, while $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ is not dominant but shares the same underlying mechanism as a background to our signal. In the inclusive analysis, the cross sections used for these two processes were not correctly configured.

To verify that these processes constitute the primary backgrounds, the $\mathcal{L} \times \sigma_{\text{hadron}}$ method is employed, where $\mathcal{L}$ is the integrated luminosity and σ_{hadron} is the hadronic cross section. The luminosity values for each data-taking period are listed in Table 1. The background yield is estimated as a function of the $\pi^+\pi^-\pi^0$ invariant mass (m) using the effective luminosity spectrum, which is related to the mass spectrum via:

$$\frac{dL_{\text{eff}}}{dm} = \frac{2m}{s} \cdot F(s, m) \cdot \mathcal{L}_{\text{int}}, \quad (1)$$

where $s = 4E^2$, E is the beam energy, $\mathcal{L}_{\text{int}}$ is the total integrated luminosity, and $F(s, m)$ is the probability function for initial-state photon emission. This function can be written as [11]:

$$\beta = \frac{2\alpha}{\pi} \left[\ln\left(\frac{s}{m_e^2}\right) - 1 \right], \quad (2)$$

$$x = 1 - \frac{m^2}{s}, \quad (3)$$

$$\begin{aligned} \frac{dL_{\text{eff}}}{dm} = \frac{2m}{s} \cdot & \left\{ \beta x^{\beta-1} \left[1 + \frac{\alpha}{\pi} \left(\frac{\pi^2}{3} - \frac{1}{2} \right) + \frac{3}{4}\beta \right. \right. \\ & \left. \left. + \beta^2 \left(\frac{37}{96} - \frac{\pi^2}{12} - \frac{1}{72} \ln \frac{s}{m_e^2} \right) \right] \right. \\ & \left. - \beta \left(1 - \frac{x}{2} \right) \ln \frac{1}{x} - \frac{1 + 3(1-x)^2}{x} \ln(1-x) - 6 + x \right\} \cdot \mathcal{L}_{\text{int}}, \quad (4) \end{aligned}$$

where $\alpha = 1/137$ is the fine-structure constant, $m_e = 0.5110034 \times 10^{-3} \text{ GeV}/c^2$ is the electron mass, and $\sqrt{s} = 3.773 \text{ GeV}$ is the center-of-mass energy of the primary e^+e^- system.

The effective luminosity spectrum calculated using this formula for round16 is shown in Fig. 2. Similar calculations were performed for all data-taking rounds.

Based on this method, rough yield estimates for the two processes with additional π^0 , $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ (non-ISR) and $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ (ISR), were calculated for the four data-taking periods (round03/04, round15, round16, round17). The results are summarized in Table 3. For the ISR channel, the estimation relies on the $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ cross section measured by the BaBar collaboration [12].

It is important to note that the values listed in the table are initial yield estimates without any selection efficiency corrections; the actual background contribution to the signal region is obtained by multiplying these yields by the corresponding selection efficiency relevant to the specific analysis cuts. These estimates serve only to confirm that these processes constitute the main sources of background and to provide guidance for optimizing subsequent selection criteria.

The final background yield is then obtained by multiplying the effective luminosity spectrum by

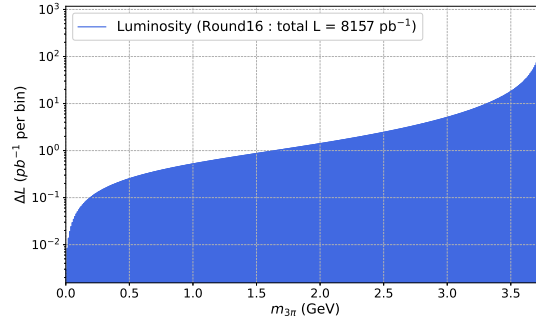


Figure 2: Effective luminosity spectrum as a function of the $\pi^+\pi^-\pi^0$ invariant mass ($m_{3\pi}$) calculated for round16 with total integrated luminosity $\mathcal{L}_{\text{int}} = 8157 \text{ pb}^{-1}$. The spectrum is shown with logarithmic scale on the vertical axis.

Table 3: Rough estimates of the expected yields for the two processes with additional π^0 before applying any selection efficiency. The integrated luminosities are taken from Table 1. The cross section for $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ is taken as $\sigma = 0.22 \text{ nb}$ at $\sqrt{s} = 3.773 \text{ GeV}$, while the energy-dependent cross section for the ISR process is derived from the measured $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ cross section [12]. The final background contribution is obtained by multiplying these yields by the corresponding selection efficiency.

Data sample (Run period)	$N(e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0)$	$N(e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}})$
round03-04	511940	1142096
round15	872209	1945825
round16	1424346	3177597
round17	731817	1632623

the cross section and a selection-efficiency factor that varies dynamically with the analysis cuts. This method is used solely to confirm that $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ and $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ are the dominant backgrounds and to optimize the selection criteria. It is not used to derive the final background estimates; precise yields for these two components will be determined using data-driven methods.

Based on the rough background estimate from $\mathcal{L} \times \sigma_{\text{hadron}}$, the signal yield was approximated by subtracting the background from the data, allowing for a preliminary significance calculation. Using this approach, a figure of merit (FOM) defined as $S/\sqrt{S+B}$ was constructed to evaluate the signal significance. To optimize the event selection, a scan over the kinematic fit χ^2 was performed, with the result shown in Fig. 3. The Round 16 optimization yields an optimal cut of $\chi_{\text{KF}}^2 < 50$.

Based on the optimal χ_{KF}^2 selection, the invariant mass spectrum of the 3π system ($M_{3\pi}$) can be divided into three distinct regions for further study: the $\omega(782)$ region, the $\phi(1020)$ region, and a higher-mass region associated with possible ω'/ω'' excitations. In the high-mass ω'/ω'' region, a significant background component from events containing additional π^0 mesons was observed. To quantify this, the number of π^0 candidates per event was counted, where any π^0 candidate satisfying a one-constraint kinematic fit $\chi^2 < 200$ was accepted. The distribution of this π^0 multiplicity is shown in Fig. 4. To suppress this specific background, a requirement of exactly one π^0 candidate per event was applied

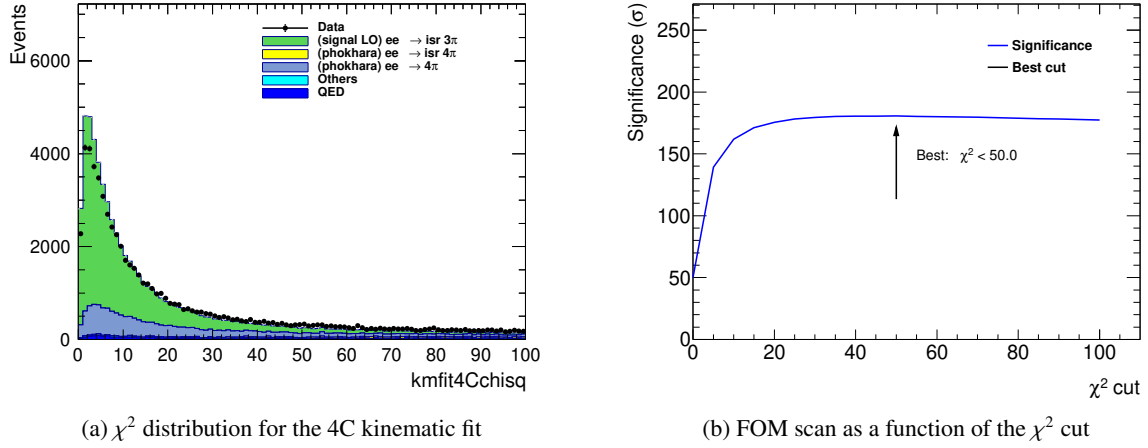


Figure 3: (a) Distribution of the kinematic fit χ^2 for the four-constraint (4C) fit using the Round 16 data sample. (b) The corresponding figure of merit ($S/\sqrt{S+B}$) as a function of the applied χ^2 cut value for the Round 16 data.

158 exclusively in the ω'/ω'' mass region. No such restriction was applied to the $\omega(782)$ and $\phi(1020)$ signal
 159 regions. The following studies and distributions are based on the Round 16 data sample.

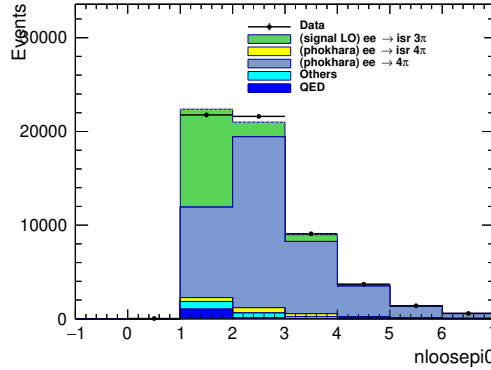


Figure 4: Distribution of the number of π^0 candidates per event in the ω'/ω'' mass region using the Round 16 data. A π^0 candidate is counted if its 1C kinematic fit $\chi^2 < 200$.

160 The resulting $M_{3\pi}$ invariant mass spectra, after applying all selection criteria including the region-
 161 specific π^0 multiplicity cut, are presented separately for the three mass regions in Fig. 5. The $e^+e^- \rightarrow$
 162 $\pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ contribution is negligible and will not be treated as a separate background in the subse-
 163 quent fitting procedure; only the non-ISR $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ process will be explicitly estimated using
 164 data-driven methods and constrained in the fit.

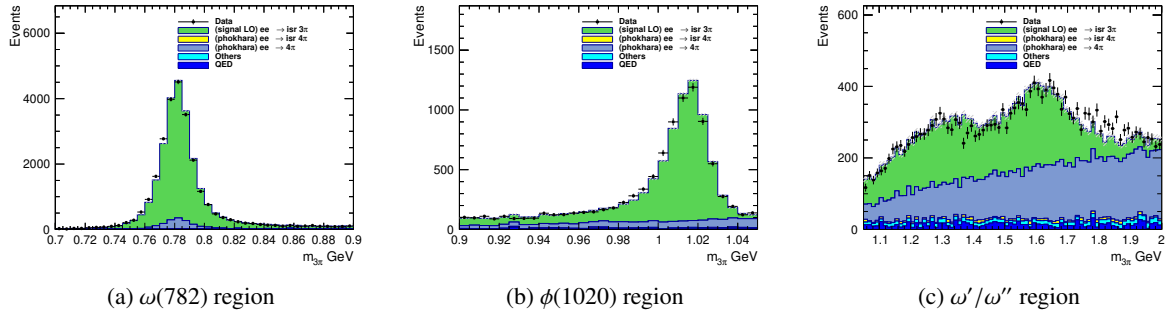


Figure 5: Invariant mass spectra $M_{3\pi}$ for the (a) $\omega(782)$, (b) $\phi(1020)$, and (c) ω'/ω'' mass regions obtained from the Round 16 data. The selection for the high-mass region (c) includes the requirement $N_{\pi^0} = 1$, while regions (a) and (b) have no π^0 multiplicity cut.

5.1 Estimate of $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$

In estimating the $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ background, the following strategy was adopted:

It is assumed that the total yield of $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ events produced in the data is a fixed quantity. This process enters our signal region primarily when a photon from one of the π^0 decays is misidentified as an initial-state radiation photon (γ_{ISR}). Consequently, two distinct selection efficiencies can be calculated: ϵ_{sig} (the efficiency for $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ events to be selected as signal candidates) and ϵ_{bkg} (the efficiency for the same process to enter the 3π signal region as background).

When $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ events are selected as signal, the same analysis procedure as for the signal is applied to construct the 3π invariant mass spectrum, thereby obtaining the distribution of this process in data. Using the ratio of these two efficiencies, $R = \epsilon_{\text{bkg}}/\epsilon_{\text{sig}}$, the observed $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ signal distribution can be converted into an estimate of its background distribution in the signal region. Specifically:

$$N_{\text{bkg}} = R \times N_{\text{sig}}^{\text{obs}} = \frac{\epsilon_{\text{bkg}}}{\epsilon_{\text{sig}}} \times N_{\text{sig}}^{\text{obs}}, \quad (5)$$

where $N_{\text{sig}}^{\text{obs}}$ is the number of observed $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ signal events in data, and N_{bkg} is the estimated number of background events from this process in the signal region.

This method provides a data-driven conversion from the observed signal distribution to the background distribution through efficiency correction.

When treating $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ as the signal, the selection criteria are designed such that the same event sample also satisfies a loose selection for $e^+e^- \rightarrow \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$. This ensures that the invariant mass of the three-pion system is automatically defined within the latter process.

The combined selection proceeds as follows: events are required to meet the criteria for both processes. The first step applies the full $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ selection:

• $\pi^+\pi^-\pi^0\pi^0$ Candidate Selection:

- Exactly two good charged tracks.
- $n_{\text{Pi0}} \geq 2$ & $n_{\text{GoodShowers}} \geq 4$ (ensuring at least two π^0 s).
- Application of a total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
- The candidate with the minimum χ_{4C}^2 is chosen.
- A requirement of $\chi_{\text{KF}}^2 < 10$ on the kinematic fit.

Subsequently, within the same event, a candidate for $e^+e^- \rightarrow \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ is formed by reinterpreting one of the π^0 s as the γ_{ISR} :

• $\gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ Candidate Selection:

- The same two good charged tracks.
- $n_{\text{Pi0}} \geq 1$ & $n_{\text{GoodShowers}} \geq 3$ (ensuring at least one π^0 and one photon assigned as γ_{ISR}).
- Application of the same total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
- The candidate with the minimum χ_{4C}^2 is chosen.

– No additional kinematic fit χ^2 requirement is applied at this stage.

This dual-tag strategy guarantees that the reconstructed 3π system from the $e^+e^- \rightarrow \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ hypothesis is kinematically well-defined, and its invariant mass can be reliably calculated for the subsequent analysis.

The selection described above results in a highly pure sample of $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ events. The invariant mass spectrum of the corresponding three-pion system from this pure sample, which serves as the input template for background estimation, is shown in Fig. 6.

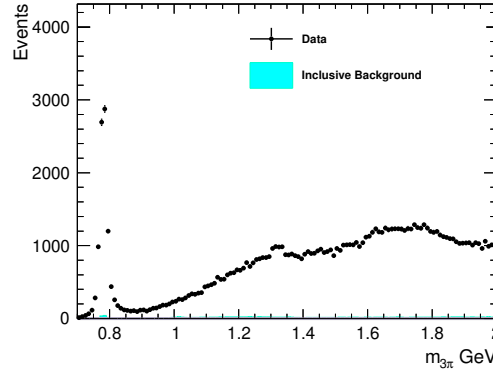


Figure 6: The 3π invariant mass spectrum from selected $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ events in the Round 16 dataset. This distribution, obtained from data after applying the dual-tag selection criteria for $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$, represents the pure signal sample used as the template for background estimation.

The observed distribution in the data, after applying the $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ selection criteria, will be scaled by the efficiency ratio $R = \epsilon_{\text{bkg}}/\epsilon_{\text{sig}}$ to estimate the final background contribution of this process in the signal region of the measurement channel. As established and shown in Fig. 6, the background contamination within this selected sample is negligible. Therefore, we subtract the inclusive background histogram from the data histogram in a symbolic manner to obtain the pure $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ event yield. The resulting background-subtracted signal distributions in the three 3π invariant mass regions are presented in Fig. 7. All other backgrounds will be subtracted directly.

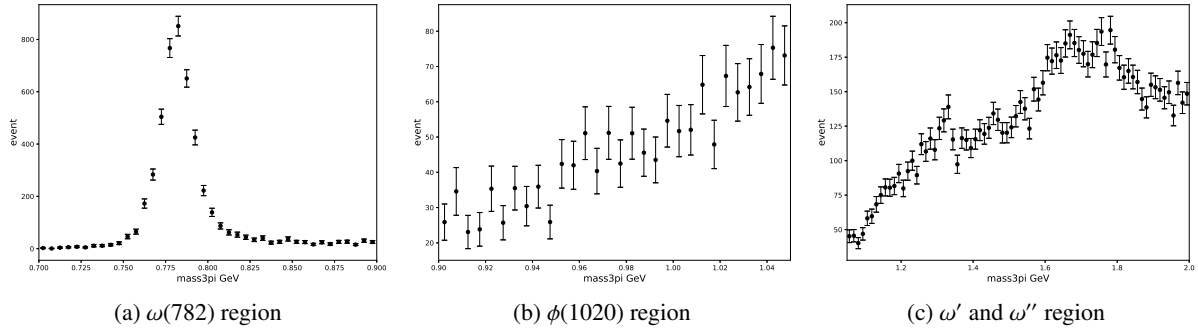


Figure 7: Estimated background contributions from $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ in the signal region after scaling, using data from round 16. The three panels correspond to different 3π invariant mass regions: (a) the $\omega(782)$ region, (b) the $\phi(1020)$ region, and (c) the higher-mass ω'/ω'' region.

5.2 Estimate of Other Backgrounds and Signal Yields

The estimation of the dominant peaking backgrounds from $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ has been completed in the previous subsection. These backgrounds exhibit shapes similar to the signal resonance peaks and are therefore categorized as peaking backgrounds.

In addition to the peaking component, a non-peaking continuum background component must also be accounted for in the analysis. The primary objective of this study is to measure the cross section for $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ via ISR. The ISR photon reduces the effective c.m. energy, allowing the measurement of the signal yield in narrow intervals of the 3π invariant mass ($M_{3\pi}$).

The $M_{3\pi}$ spectrum is divided into three physically motivated regions, each subdivided into multiple bins for detailed study:

- The $\omega(782)$ region (0.7–0.9 GeV/ c^2), divided into 40 bins.
- The $\phi(1020)$ region (0.9–1.05 GeV/ c^2), divided into 30 bins.
- The ω'/ω'' region (1.05–2.0 GeV/ c^2), divided into 76 bins.

For each small $M_{3\pi}$ interval, the signal yield is extracted through an extended unbinned maximum-likelihood fit to the U_{miss} distribution, where U_{miss} is defined as the difference between the missing energy and the magnitude of the missing momentum ($U_{\text{miss}} = E_{\text{miss}} - |\vec{p}_{\text{miss}}|$). The fit model incorporates probability density functions (PDFs) representing the signal and all background components.

To illustrate the composition of U_{miss} distributions across different $M_{3\pi}$ regions, the inclusive MC simulation is examined. Figure 8 shows the U_{miss} distributions for the full $\omega(782)$ region, the full $\phi(1020)$ region, and the full ω'/ω'' region respectively, with the peaking background contribution from $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ stacked according to its expected yield. These distributions demonstrate the characteristic features of each mass region that motivate our fitting strategy.

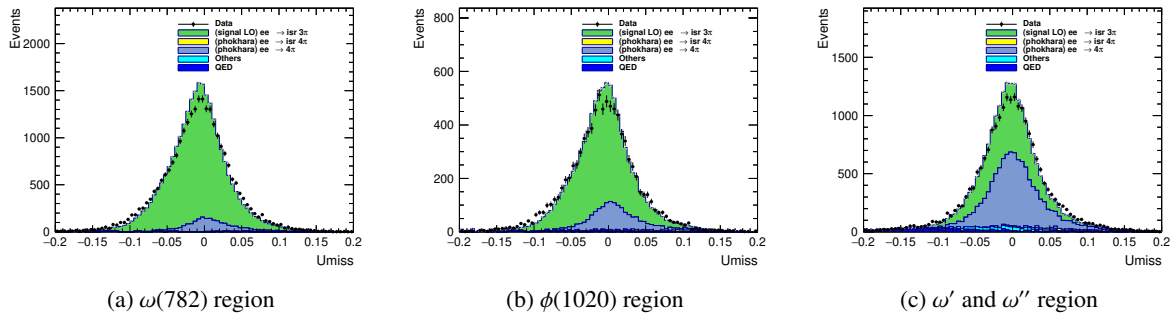


Figure 8: U_{miss} distributions from inclusive MC simulation for three $M_{3\pi}$ regions: (a) the $\omega(782)$ region, (b) the $\phi(1020)$ region, and (c) the higher-mass ω'/ω'' region.

Based on the understanding from the inclusive MC, the PDFs for the fit are constructed as follows:

- **Signal PDF:** Histogram shape from signal MC, convolved with a Gaussian for resolution correction.

- **Peaking background PDF:** Shape for $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ from dedicated MC, convolved with the same Gaussian as signal. Yield constrained by Gaussian term using data-driven estimate from Section 5.
- **PDF shape sharing across bins:** Limited MC statistics preclude bin-by-bin shape extraction. Instead:
 - For the $\omega(782)$ and $\phi(1020)$ regions, where the $M_{3\pi}$ bin width is $5 \text{ MeV}/c^2$, the PDF shapes are extracted from MC simulation every five consecutive bins, corresponding to $25 \text{ MeV}/c^2$ intervals. All five bins within each such interval share the same PDF shape, under the assumption that the U_{miss} distribution varies slowly over this mass range.
 - For the ω'/ω'' region, where the $M_{3\pi}$ bin width is $12.5 \text{ MeV}/c^2$, the PDF shapes are also extracted every $25 \text{ MeV}/c^2$. Each extracted shape is therefore shared by two adjacent bins, balancing the need for shape precision against the limited MC statistics in this higher-mass region.
- **Continuum background PDF:** All remaining minor background processes are collectively described by a polynomial function. A first-order polynomial (linear function) is used for the $\omega(782)$ and $\phi(1020)$ regions, while a second-order Chebyshev polynomial is employed for the ω'/ω'' region, as suggested by the shape of the inclusive MC distribution in Fig. 8c.

The fit procedure is validated by examining the quality of the fits to data in representative $M_{3\pi}$ bins across the three mass regions. Figure 9 presents typical fit results for three selected bins, one from each region, demonstrating the ability of the fit model to accurately describe the data. The signal component, peaking background from $\pi^+\pi^-\pi^0\pi^0$, and polynomial continuum background are clearly distinguished, and the total PDF provides an excellent description of the data distribution in each case.

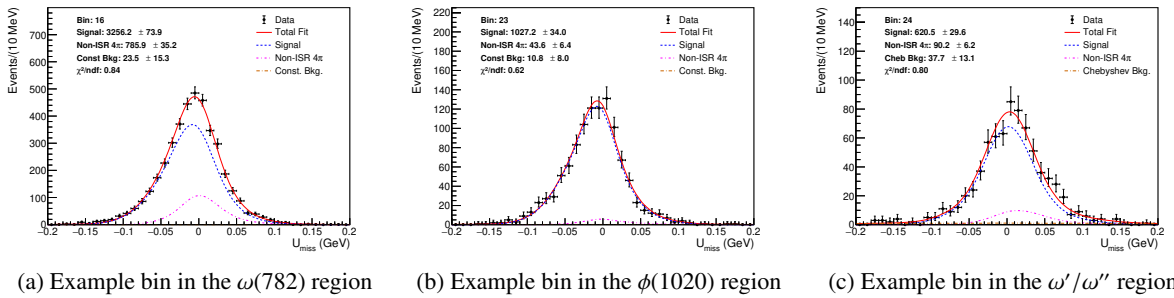


Figure 9: Typical U_{miss} fit results for representative $M_{3\pi}$ bins from each of the three mass regions. The data points are shown with error bars, the solid blue curve represents the total fit PDF, the red dashed curve shows the signal component, the green shaded area indicates the peaking background from $\pi^+\pi^-\pi^0\pi^0$, and the magenta dashed line represents the polynomial continuum background.

Applying this fitting procedure to all $M_{3\pi}$ bins yields the extracted signal yields across the entire mass spectrum. Figure 10 shows the measured signal yields as a function of $M_{3\pi}$ for the three regions. The $\omega(782)$ resonance is clearly visible as a prominent peak around $0.78 \text{ GeV}/c^2$, while the $\phi(1020)$

264 resonance appears as a distinct but smaller peak near $1.02 \text{ GeV}/c^2$. In the higher mass region, broad
 265 structures corresponding to the ω' and ω'' excitations are observed, with significantly reduced yields due
 266 to the decreasing cross section and lower ISR photon probability at larger momentum transfers.

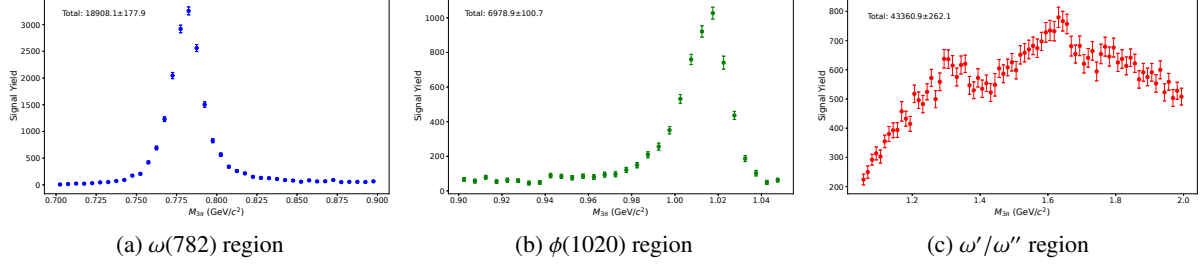


Figure 10: Extracted signal yields as a function of $M_{3\pi}$ for the three mass regions. The error bars represent the statistical uncertainties from the fits. The $\omega(782)$ and $\phi(1020)$ resonances appear as clear peaks, while the higher-mass region reveals broader structures corresponding to excited ω states.

267 The signal yields obtained from this fitting procedure provide the foundation for calculating the
 268 Born cross section for $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ as a function of $M_{3\pi}$. After applying efficiency corrections,
 269 unfolding for resolution effects, and subtracting residual background contributions, the cross section will
 270 be determined and compared with previous measurements and theoretical predictions.

6 Unfolding

To extract the dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$, we need the generator-level yield $\left(\frac{dN}{dm}\right)^{\text{gen}}$, which is related to the cross section within each mass interval by:

$$\left(\frac{dN}{dm}\right)^{\text{gen}} = \sigma_{3\pi}^{\text{dress}}(m) \cdot \frac{dL}{dm} \cdot \varepsilon \cdot \mathcal{B}(\pi^0 \rightarrow \gamma\gamma) \quad (6)$$

where $\frac{dL}{dm}$ is the integrated luminosity in that interval, and $\mathcal{B}(\pi^0 \rightarrow \gamma\gamma)$ is the branching fraction of the $\pi^0 \rightarrow \gamma\gamma$ decay.

From data, we have obtained the observed signal yield $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$ in each mass interval by fitting the U_{miss} distribution, as shown in Fig. 10. Due to detector effects, events can migrate between different mass intervals, so these observed yields do not directly give the generator-level yields in Eq. (6). The migration is described by the response matrix P_{ij} , which relates the observed yield in bin i to the generator-level yield in bin j :

$$\left(\frac{dN}{dm}\right)_i^{\text{obs}} = \sum_j P_{ij} \left(\frac{dN}{dm}\right)_j^{\text{gen}} \quad (7)$$

Let A_{ij} denote the number of MC events that are generated in true bin j and reconstructed in bin i , where i and j index the observed and generator-level mass bins, respectively. The migration matrices A_{ij} for three representative mass regions are shown in Fig. 11, illustrating how events generated in a given true bin are distributed across observed bins due to detector effects. This matrix is then normalized column-wise to obtain the response matrix used in Eq. (7):

$$P_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_d} A_{kj}}, \quad \sum_{i=1}^{n_d} P_{ij} = 1, \quad (8)$$

where n_d is the number of reconstructed bins.

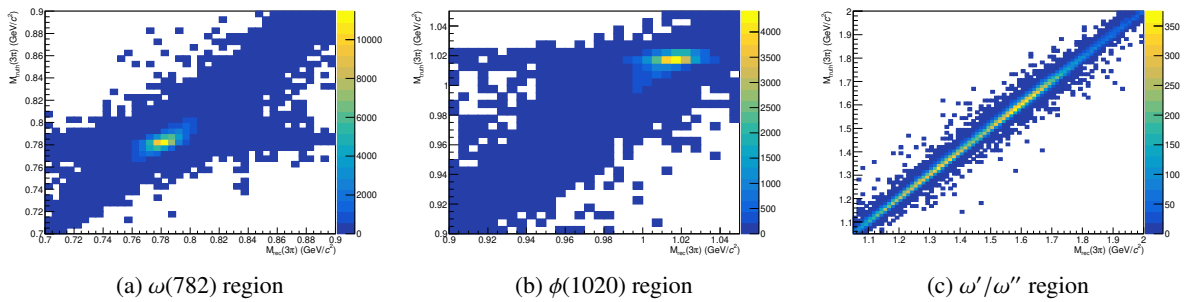


Figure 11: Response matrices A_{ij} for three representative mass regions, showing the migration of events between generator-level and reconstructed $M_{3\pi}$ bins. The diagonal structure indicates good resolution, while off-diagonal entries reflect the effects of finite detector resolution and acceptance.

6.1 Unfolding the generator-level yield with IDS

To extract the generator-level yield $\left(\frac{dN}{dm}\right)_j^{\text{gen}}$ from the observed yields $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$ and the response matrix P_{ij} , we employ the Iterative Dynamically Stabilized (IDS) unfolding method [?]. This section introduces the mathematical formulation of the method; the specific implementation, parameter optimization, and iteration strategy are described in the following sections.

From the MC simulation, we have constructed the migration matrix A_{ij} (shown in Fig. 11) and the response matrix P_{ij} normalized column-wise as in Eq. (8). The corresponding unfolding matrix is defined row-wise:

$$\tilde{P}_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_u} A_{ik}}, \quad (9)$$

which gives the probability that an event reconstructed in bin i originated from true bin j .

The IDS method uses a regularization function $f(|\Delta|, \tilde{\sigma}, \lambda)$ that dynamically distinguishes statistical fluctuations from genuine signals. Here, $|\Delta|$ represents the absolute difference between data and the model prediction (e.g., $|\Delta d_i|$ for the discrepancy in reconstructed bin i , or $|\Delta u_j|$ for the discrepancy in true bin j), $\tilde{\sigma}$ denotes the corresponding statistical uncertainty ($\tilde{\sigma} d_i$ or $\tilde{\sigma} u_j$), and λ is a tunable regularization parameter. The function satisfies the following asymptotic behavior:

$$\lim_{\frac{|\Delta|}{\lambda \tilde{\sigma}} \rightarrow 0} f(|\Delta|, \tilde{\sigma}, \lambda) = 0, \quad \lim_{\frac{|\Delta|}{\lambda \tilde{\sigma}} \rightarrow \infty} f(|\Delta|, \tilde{\sigma}, \lambda) = 1. \quad (10)$$

Thus, for discrepancies much smaller than λ times the uncertainty, the function approaches zero, treating the difference as a fluctuation; for discrepancies much larger, it approaches unity, indicating a real signal that should be unfolded. The transition between these regimes is smooth, avoiding sharp cutoffs that could introduce artifacts.

In the IDS method, the unfolding is performed iteratively. At iteration k , the unfolded spectrum $u_j^{(k)}$ is given by:

$$u_j^{(k)} = \eta \cdot t_j + \sum_{i=1}^{n_d} \left[f(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U) \Delta d_i^{(k-1)} \tilde{P}_{ij}^{(k-1)} + \left(1 - f(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U)\right) \Delta d_i^{(k-1)} \delta_{ij} \right], \quad (11)$$

where:

- $u_j^{(k)}$ is the unfolded generator-level yield in bin j at iteration k ,
- t_j is the true MC yield in bin j from simulation,
- $\eta = N_d^{\text{MC}}/N_{\text{MC}}$ is the normalization factor that scales the MC to match the data in regions without significant new structures,
- $\Delta d_i^{(k-1)} = d_i - B_i^d - \eta \cdot r_i^{(k-1)}$ is the difference between the observed data yield d_i and the sum of the estimated background B_i^d (which may be zero if no background subtraction is applied) and the normalized reconstructed MC spectrum $\eta \cdot r_i^{(k-1)}$,
- $r_i^{(k-1)}$ is the reconstructed MC spectrum at iteration $k-1$, obtained by folding the unfolded spectrum $u_j^{(k-1)}$ with the response matrix $P_{ij}^{(k-1)}$,

- $\tilde{\sigma}d_i$ is the statistical uncertainty of d_i ,
- $\tilde{P}_{ij}^{(k-1)}$ is the unfolding matrix from the previous iteration,
- λ_U is the regularization parameter used in the unfolding step,
- δ_{ij} is the Kronecker delta, which equals 1 if $i = j$ and 0 otherwise.

The first term on the right-hand side, $\eta \cdot t_j$, represents the normalized true MC spectrum, which serves as a baseline. The remaining sum dynamically partitions the discrepancy $\Delta d_i^{(k-1)}$: a fraction f is unfolded using $\tilde{P}_{ij}^{(k-1)}$, while the complementary fraction $1 - f$ is retained in the original bin $j = i$ via the Kronecker delta.

The response matrix itself can be iteratively improved. Using the difference $\Delta u_j = u_j - \eta \cdot t_j$ between the current unfolded result and the normalized true MC, the migration matrix is modified as:

$$A'_{ij} = A_{ij} + f \left(|\Delta u_j|, \tilde{\sigma}u_j, \lambda_M \right) \Delta u_j P_{ij} \frac{N_{MC}}{N_d^{MC}}, \quad \text{for } i \in \{1, \dots, n_d\}, \quad (12)$$

where:

- Δu_j is the difference between the current unfolded yield u_j and the normalized true MC $\eta \cdot t_j$,
- $\tilde{\sigma}u_j$ is the statistical uncertainty associated with u_j ,
- λ_M is the regularization parameter controlling the modification strength,
- P_{ij} is the current response matrix (column-normalized A_{ij}),
- $N_{MC} = \sum_j t_j$ is the total number of generated MC events,
- N_d^{MC} is the number of data events compatible with the MC model (used for normalization).

The updated migration matrix A'_{ij} is then renormalized column-wise to obtain an improved response matrix P_{ij} , and row-wise to obtain an improved unfolding matrix $\tilde{P}_{ij}$.

6.2 Parameter optimization and figure of merit

The IDS method involves three regularization parameters: λ_L for the initial unfolding, λ_M for modifying the response matrix, and λ_U for subsequent unfoldings. The iterative procedure consists of the following three steps:

1. **Initial unfolding:** Perform an unfolding using Eq. (11) with $\lambda = \lambda_L$ and the initial response matrix P_{ij} , obtaining $u_j^{(0)}$.
2. **Matrix improvement:** Modify the migration matrix via Eq. (12) using λ_M and the current u_j . Renormalize to obtain updated P_{ij} and $\tilde{P}_{ij}$.
3. **Improved unfolding:** Perform an unfolding using Eq. (11) with $\lambda = \lambda_U$ and the updated matrices.

These three steps can be repeated for a chosen number of iterations.

In this analysis, the four parameters λ_M , λ_L , λ_U , and n_{iter} are optimized by scanning over a grid of possible values. For each parameter combination $(\lambda_M, \lambda_L, \lambda_U, n_{\text{iter}})$, we perform the IDS unfolding and evaluate the quality of the result using a χ^2 figure of merit. The χ^2 is computed by comparing the unfolded spectrum u_j to the true generator-level distribution t_j^{truth} from MC simulation. The MC sample used for this comparison has been normalized such that its total integral matches the total number of signal events observed in data (shown in Fig. 10), ensuring that the statistical uncertainties from the toy ensembles properly reflect the data statistics.

$$\chi^2 = (\mathbf{u} - \mathbf{t}^{\text{truth}})^T \Sigma^{-1} (\mathbf{u} - \mathbf{t}^{\text{truth}}), \quad (13)$$

where $\mathbf{u}$ and $\mathbf{t}^{\text{truth}}$ are vectors of bin contents, and Σ is the covariance matrix of the unfolded spectrum. The covariance matrix accounts for the statistical uncertainties and their correlations introduced by the unfolding procedure. It is estimated using a MC sampling method:

1. Generate an ensemble of N_{toy} pseudo-data distributions by fluctuating the original observed spectrum according to its statistical uncertainties, using a random number generator.
2. For each pseudo-data distribution, perform the IDS unfolding with the same parameter combination $(\lambda_L, \lambda_U, n_{\text{iter}})$, obtaining an unfolded spectrum $u_j^{(k)}$ for toy k .
3. From the ensemble of unfolded spectra, compute the covariance matrix:

$$\Sigma_{ij} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} (u_i^{(k)} - \bar{u}_i) (u_j^{(k)} - \bar{u}_j), \quad (14)$$

where $\bar{u}_i = \frac{1}{N_{\text{toy}}} \sum_k u_i^{(k)}$ is the mean unfolded yield in bin i across the toy ensemble.

In cases where the covariance matrix is singular or ill-conditioned, a small diagonal perturbation $\lambda_{\text{reg}} I$ is added before inversion to ensure numerical stability, with $\lambda_{\text{reg}} = 10^{-6}$ in this analysis.

The optimal parameter combination is chosen based on several considerations: the number of iterations n_{iter} should be as small as possible to avoid unnecessary complexity; the χ^2 value defined in Eq. (13) should be small, indicating good agreement between the unfolded and true spectra; and the χ^2 per degree of freedom, χ^2/n_{dof} , should be close to or less than one, indicating that the uncertainties are well estimated. Using the MC sample as a stand-in for data, a scan is performed over a grid of λ_L , λ_U , and n_{iter} values, and the combination that best satisfies these criteria is selected for the final unfolding of the data. The optimized parameters obtained from this procedure for the three representative mass regions are summarized in Table 4. The unfolding results obtained with these optimized parameters are shown in Fig. 12.

To validate the unfolding procedure, we compare the relative statistical uncertainty of the unfolded spectrum with that of the test MC sample. The relative uncertainty of the test MC spectrum, δ_{test} , is computed by summing in quadrature the Poisson errors of each bin and dividing by the total yield:

$$\delta_{\text{test}} = \frac{\sqrt{\sum_i \sigma_i^2}}{\sum_i N_i}, \quad (15)$$

Table 4: Optimized unfolding parameters for the three representative mass regions. For each region, the parameters λ_L , λ_U , λ_M , and number of iterations n_{iter} were selected based on the χ^2 minimization procedure described in the text. The resulting χ^2/ndf values indicate good agreement between the unfolded and true MC distributions.

Region	λ_L	λ_U	λ_M	n_{iter}	χ^2/ndf
$\omega(782)$	1×10^{-3}	1×10^{-4}	1	4	28.31/40
$\phi(1020)$	1×10^{-4}	1×10^{-4}	1×10^{-3}	4	23.86/30
ω'/ω''	1×10^{-3}	1	11	5	76.18/76

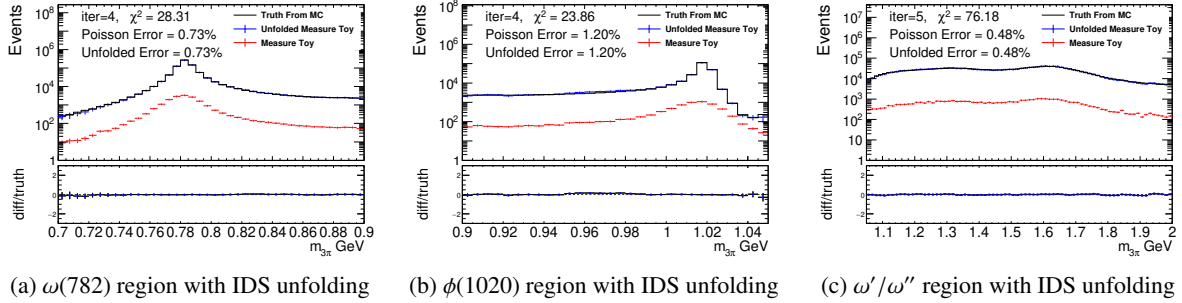


Figure 12: Unfolded results for three representative mass regions using the IDS method with optimized parameters. Each panel shows the true MC distribution (black), the measured MC distribution (red), and the unfolded result (blue points) with its statistical uncertainties. The MC sample has been normalized to match the data statistics shown in Fig. 10, ensuring that the uncertainties properly reflect the data statistical power.

where N_i and σ_i are the content and Poisson error of bin i in the test MC distribution, respectively. This represents the Poisson statistical uncertainty of the test MC sample.

The relative uncertainty of the unfolded spectrum, δ_{unfolded} , is derived from the full covariance matrix Σ obtained from the toy MC ensemble:

$$\delta_{\text{unfolded}} = \frac{\sqrt{\sum_{i,j} \Sigma_{ij}}}{\sum_i u_i}, \quad (16)$$

where u_i are the bin contents of the unfolded spectrum. The relative uncertainty of the unfolded spectrum should not be smaller than the Poisson uncertainty of the test MC sample.

382 **7 Efficiency correction**

383 to fill

384 **8 Systematic uncertainties**

385 to fill

386 **9 Summary**

387 to fill

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404 **APPENDIX**